

RetinaScan AI

Clinical Validation Report — External Dataset

Key Performance Indicators

77.2%

Overall Accuracy

95% CI: 75.7% – 78.5%

98.2%

Referable DR Sensitivity

95% CI: 97.5% – 98.9%

86.5%

Referable DR Specificity

0.934

AUC (Referable DR)

Dataset

External dataset: APTOS 2019 Blindness Detection (Indian diabetic retinopathy dataset)

Total images: 3662 (all unseen during model training)

Evaluated: 3394 | **Rejected by IQA:** 268 (7.3%)

Model: EfficientNet-B2 (ONNX Runtime, CPU)

Preprocessing: CLAHE → green channel → pupil-centered crop → 512×512

Per-Class Performance

Class	Precision	Recall	F1	Support
No DR	0.974	0.985	0.979	1605
Mild NPDR	0.765	0.216	0.337	361
Moderate NPDR	0.595	0.942	0.729	968
Severe NPDR	0.361	0.265	0.306	181
Proliferative DR	1.000	0.004	0.007	279

Confusion Matrix

True \ Pred	No DR	Mild	Mod	Sev	PDR
No DR	1581	13	11	0	0
Mild	29	78	252	2	0
Mod	11	11	912	34	0
Sev	1	0	132	48	0
PDR	2	0	227	49	1

Clinical Significance

This validation demonstrates that RetinaScan AI achieves **98.2% sensitivity** for detecting referable diabetic retinopathy (ICDR Stage ≥ 2) on an external dataset of **3394 unseen images**. At the operating point of 86.5% specificity, the system would correctly identify approximately 98% of patients requiring ophthalmologist referral while flagging only 13% false positives for manual review.

These results support the use of RetinaScan AI as a **screening triage tool** in settings with limited ophthalmologist availability, particularly in rural and underserved communities where diabetic retinopathy screening coverage is lowest.

This report was generated by an automated validation pipeline. Results are on a single external dataset and do not constitute clinical validation for regulatory purposes. Further multi-site, prospective studies are required.